



Development and external validation of a machine learning-based model to classify uric acid stones in patients with kidney stones of Hounsfield units < 800

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Abstract

The correct diagnosis of uric acid (UA) stones has important clinical implications since patients with a high risk of perioperative morbidity may be spared surgical intervention and be offered alkalization therapy. We developed and validated a machine learning (ML)-based model to identify stones on computed tomography (CT) images and simultaneously classify UA stones from non-UA stones. An international, multicenter study was performed on 202 patients who received percutaneous nephrolithotomy for kidney stones with HU < 800. Data from 156 (77.2%) patients were used for model development, while data from 46 (22.8%) patients from a multinational institution were used for external validation. A total of 21,074 kidney and stone contour-annotated CT images were trained with the ResNet-18 Mask R-convolutional neural network algorithm. Finally, this model was concatenated with demographic and clinical data as a fully connected layer for stone classification. Our model was 100% sensitive in detecting kidney stones in each patient, and the delineation of kidney and stone contours was precise within clinically acceptable ranges. The development model provided an accuracy of 99.9%, with 100.0% sensitivity and 98.9% specificity, in distinguishing UA from non-UA stones. On external validation, the model performed with an accuracy of 97.1%, with 89.4% sensitivity and 98.6% specificity. SHAP plots revealed stone density, diabetes mellitus, and urinary pH as the most important features for classification. Our ML-based model accurately identified and delineated kidney stones and classified UA stones from non-UA stones with the highest predictive accuracy reported to date. Our model can be reliably used to select candidates for an earlier-directed alkalization therapy.

Keywords Decision support techniques · Machine learning · Urolithiasis · Validation

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Introduction

The selection of a treatment modality for patients with kidney stones is based on stone size, location, density, instrument availability, and patient comorbidities. This choice affects postoperative outcomes and perioperative morbidity. Treatment modalities recommended by contemporary guidelines include ureteroscopic lithotripsy, percutaneous nephrolithotripsy (PCNL), extracorporeal shockwave lithotripsy, and alkalization therapy [1–3]. For selected stones, surgical treatment options may be equally effective regarding the outcome. Therefore, a specific treatment modality can be recommended based on the physician's expertise and patient preference [1]. On the other hand, surgical intervention may compromise perioperative outcomes and increase morbidity in patients with significant comorbidities. In some patients, non-surgical modalities that are less invasive may be beneficial.

Uric acid (UA) stones comprise 5 to 20% of all urolithiasis and are primarily explained by a low urinary pH, with a minority of patients exhibiting high urinary excretion of UA [4, 5]. Medical dissolution therapy with urinary alkalization is the cornerstone of the medical management of UA stones, with a reported success rate of 80% and cost savings compared to surgical intervention [5]. The prediction of UA stones is not entirely accurate and is based on Hounsfield units (HU) < 450 on non-contrast-enhanced computed tomography (NCCT), radiolucency on plain radiography, and low urinary pH < 6 [6]. However, UA stones outside these ranges are often underdiagnosed due to a lack of reliable diagnostic systems predicting UA components and concerns about stones mistakenly thought to be UA [7]. As a result, alkalization therapy is reported to be underused despite its potential benefits [6]. Correct differential diagnosis of UA stones from other stone components has important clinical implications since patients with a high risk of perioperative morbidity may be spared surgical interventions.

Several non-invasive techniques have been developed to classify UA from non-UA stones, including 24-h urinalysis, nomograms using clinical and HU parameters, and dual-energy CT (DECT) [8–10]. However, these classification systems were limited by their model developments being based on a specific stone component, the necessity of access to particular CT scanner types, and, most importantly, the lack of external validation, which precludes their general applicability to clinical use. A growing body of evidence indicates that machine learning (ML) models may improve the accuracy of disease diagnosis and treatment outcomes compared to conventional discriminant analyses [11]. Given the multifactorial nature of the formation of specific stone components, we sought to develop

an ML-based model that incorporates a comprehensive set of demographical, clinical, and CT data to better classify UA stones.

The aims of our study were (1) to train and develop an ML-based urinary stone recognition algorithm that can automatically identify the stone location and delineate its contour relative to the kidney and (2) to utilize this algorithm to develop a prediction model that incorporates demographic, clinical, and NCCT data to classify UA stones from other stone components.

Materials and methods

Study cohort

An international, multicenter, cross-sectional study was conducted on 242 patients who underwent PCNL for kidney stones with HU < 800 between March 2005 and November 2018. The development cohort consisted of 156 (77.2%) patients from the Stone Centre at Vancouver General Hospital in Canada, while the external validation cohort comprised 46 (22.8%) patients from Gangnam Severance Hospital in Seoul, South Korea, a multinational institution with patients of a distinct ethnic background. Patients with mixed component stones ($n = 21$; 8.7%) and incomplete data, including unknown stone composition ($n = 5$; 2.5%), were excluded from the analysis. In case a patient received multiple procedures, information obtained after the first procedure was also excluded ($n = 14$; 6.9%).

This study was approved by the institutional ethics committees of the University of British Columbia's Clinical Research Ethics Board (H14-00475) and Gangnam Severance Hospital (2019-0838-001) after reviewing the protocols employed. All study procedures complied with the principles of the 1946 Declaration of Helsinki and its 2008 update.

Data acquisition

The demographic and clinical data that potentially affect stone components and NCCT images were retrospectively collected from the patients' medical records. Twenty-six preoperative demographic and clinical data included the following: patient age, body mass index (BMI), gender, American Society of Anesthesiologists score, stone history, number, bilaterality, multiplicity, HU, urinary pH and nitrite positivity, urine culture, serum levels of sodium, potassium, calcium, glomerular filtration rate, UA, phosphate, and the presence of comorbidities including cerebrovascular disease, chronic kidney disease defined as glomerular filtration rate < 60 mL/min/1.73 m², chronic obstructive pulmonary disease, diabetes mellitus, hypertension, gout, and neurological disease prior to PCNL (Table 1). Stone component data

Table 1 Demographic and clinical characteristics of the development and external validation cohorts

	Development cohort	External validation cohort	<i>P</i>
No.	156 (77.2%)	46 (22.8%)	NS
Age (year)	59.0 (47.0–68.3)	54.0 (37.0–64.5)	0.030
Body mass index (kg/m ²)	29.1 (24.3–33.3)	24.8 (21.8–27.6)	<0.001
Gender			0.467
Male	101 (64.7%)	26 (56.5%)	
Female	55 (35.3%)	20 (43.5%)	
Hounsfield units	499.4 (413.6–629.3)	545.5 (487.3–699.5)	<0.001
Stone, number	3.0 (2.0–4.5)	4.0 (3.0–6.5)	0.065
Stone, surface area (mm ²)	286.0 (37.0–676.0)	340.0 (202.0–606.0)	0.251
Stone component			
UA	41 (26.3%)	13 (28.3%)	0.143
Struvite	56 (35.9%)	14 (30.4%)	0.508
Calcium oxalate	54 (34.6%)	17 (37.0%)	0.236
Cystine	5 (3.2%)	2 (4.3%)	0.987
Urinary pH	5.5 (5.0–6.5)	6.5 (5.5–7.3)	0.003
Nitrite positivity	32 (20.5%)	9 (19.6%)	0.566
Urine culture positivity	31 (19.9%)	8 (17.4%)	0.648
Serum			
Sodium	140.0 (139.0–142.0)	140.0 (140.0–141.0)	0.587
Potassium	4.1 (3.8–4.4)	4.20 (4.10–4.38)	0.864
Calcium	2.33 (2.26–2.42)	2.31 (2.24–2.47)	0.873
Glomerular filtration rate	69.0 (50.0–94.0)	90.0 (74.0–121.5)	<0.001
UA	339.5 (273.5–431.5)	298.0 (206.5–350.5)	0.018
Phosphate	1.10 (0.92–1.29)	1.06 (0.96–1.28)	0.928
Bilaterality	42 (26.9%)	16 (34.8%)	0.040
Multiplicity	89 (57.1%)	37 (80.4%)	0.009
ASA score			0.062
0	38 (24.4%)	9 (19.6%)	
≥ 1	118 (75.6%)	37 (80.4%)	
Stone history	40 (25.6%)	10 (21.7%)	0.098
Comorbidity			
Cerebrovascular disease	18 (11.5%)	5 (10.9%)	0.438
Chronic kidney disease	12 (7.7%)	3 (6.5%)	0.768
Chronic obstructive pulmonary disease	10 (6.4%)	2 (4.3%)	0.133
Diabetes mellitus	45 (28.8%)	10 (21.7%)	0.389
Hypertension	54 (34.6%)	20 (43.5%)	0.164
Gout	3 (1.9%)	0 (0.0%)	0.418
Neurological disease	20 (12.8%)	5 (10.9%)	0.129

Values are presented as number (%) or median (interquartile range)

ASA American Society of Anesthesiologists, COPD chronic obstructive pulmonary disease, UA uric acid

for both the development and external validation cohorts were based on analyses performed using Fournier-transform infrared spectrometry, carried out at Lifelabs, Burnaby, BC, Canada, and Green Cross Laboratories, Yongin, South Korea, respectively.

Standard axial NCCT images that included kidney and stone information were acquired in DICOM format. After acquisition, the images were extracted using a Python

software application and saved in PNG files. A total of 14,843 and 6231 kidney and stone images were selected to form the datasets for developing the ML model. The anatomical contours of the kidneys and stones were semi-automatically annotated using the open-source Computer Vision Annotation Tool (Intel®, CA, USA). Overall, 21,074 kidney and stone annotated images were included in the datasets for model development.

Model development

Figure 1 depicts the overall architecture of our model. Initially, the ResNet18 model framework was trained with the PyTorch feedforward deep learning library using predefined methods [12]. We used stochastic gradient descent as the optimizer with a learning rate set at 0.005 and a batch size of 32. The ResNet-18 Mask R-convolutional neural network model was chosen for this study due to its capability and robustness in general-purpose image segmentation under limited data [13]. The kidney and stone contour-annotated image data from the development cohort was used for model training, validation, and testing in an 8:1:1 ratio, with all images in the training set being different from those in the testing set. Finally, demographic and clinical data were concatenated with the ResNet18 model in a fully connected layer to develop the final model for stone component classification. The model was then interpreted using the SHAP algorithm to enable visual interpretation of the quantitative association between the input variables and the model's output [14].

Evaluation metrics

Prediction accuracies were analyzed according to binary (UA vs. non-UA stones) and multiclass (UA vs. calcium oxalate, struvite, and cystine stones) classifications.

Prediction accuracies were measured by the precision of instance prediction, which counted the number of true positive, true negative, false positive, and false negative instances as compared with the operator's semi-automatic annotation. Prediction accuracies were compared to those of the multivariate logistic regression analyses.

Statistical analysis

Demographic and clinical characteristics between the development and external validation cohorts were compared using the two-sided Mann–Whitney *U* test for the analysis of continuous variables and the chi-square test for the analysis of categorical variables. Logistic regression analysis for binary classification was performed using the same training set. Independent predictive indicators associated with UA stones in the multivariate analyses were entered into the logistic regression model. All tests were two-tailed, with statistical significance set at a $p < 0.05$. Statistical analysis was performed using IBM SPSS Statistics software ver. 21.0 (IBM Corporation, Armonk, NY) and R Statistical Package ver. 3.1.3. (Institute for Statistics and Mathematics, Vienna, Austria).

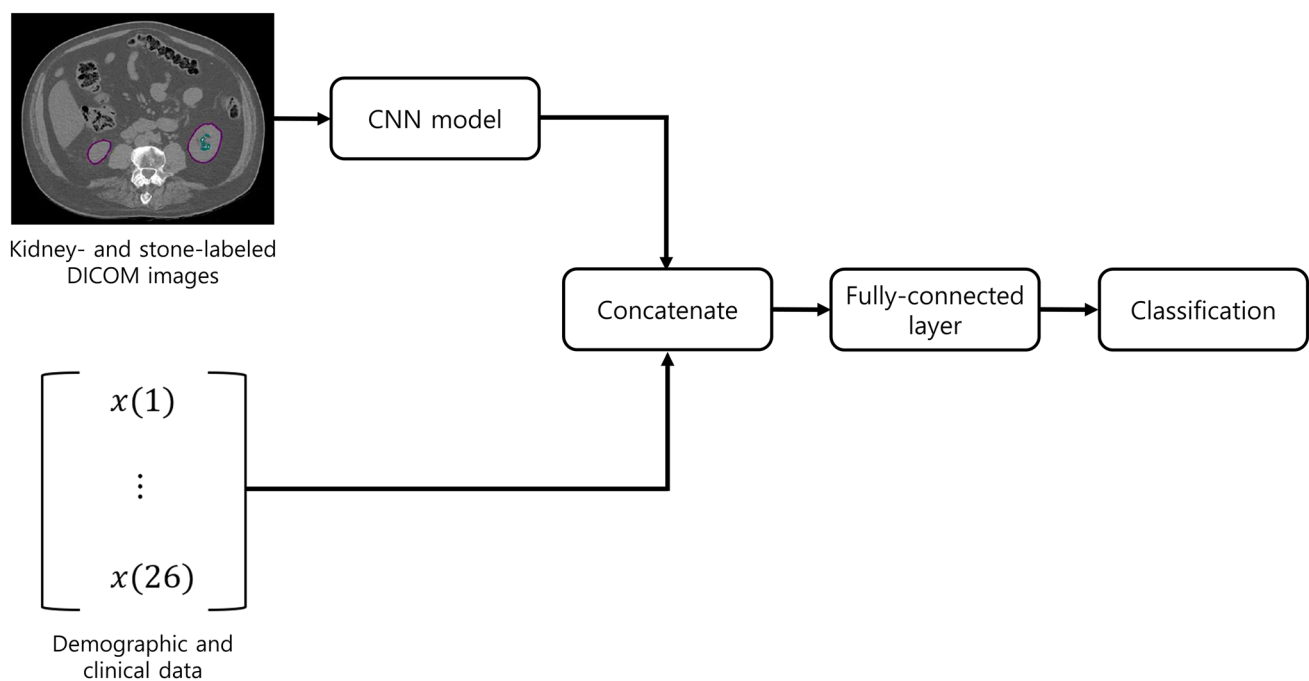


Fig. 1 Overall architecture of our model: first, the ResNet18 Mask R-convolutional neural network model framework was trained using predefined methods with the PyTorch feedforward deep learning

library. Next, demographic and clinical data were concatenated with the ResNet18 model in a fully connected layer to develop the final model for stone component classification

Results

Demographic and clinical features

The demographic and clinical characteristics of 202 patients in the development and external validation cohorts are presented in Table 1. According to stone components, UA, calcium oxalate, struvite, and cystine stones comprised 26.3%, 35.9%, 34.6%, and 3.2% in the development cohort, while 28.3%, 30.4%, 37.0%, and 4.3% in the external validation cohort, respectively. Patients in the development cohort had older age, higher BMI and serum UA levels, and lower proportions of stone multiplicity, bilaterality, and HU stones, and lower levels of urinary pH and glomerular filtration rate compared to the external validation cohort.

Kidney and stone identification and contour delineation

Segmentation identified and delineated two anatomical elements, kidney and stone contours. Our model was 100% sensitive in detecting kidney stones in each patient. The delineation of kidney and stone contours was precise within clinically acceptable ranges, as shown in a selected sample patient (Fig. 2).

Predictive performance

The predictive accuracies of our model varied according to the stone component and classification system. Overall, the performances of the ML-based model outperformed those of the logistic regression model (Table 2).

Fig. 2 Patient-specific human-annotated and model-interpreted kidney and stone contour segment of a chosen sample patient (subject no. 78)

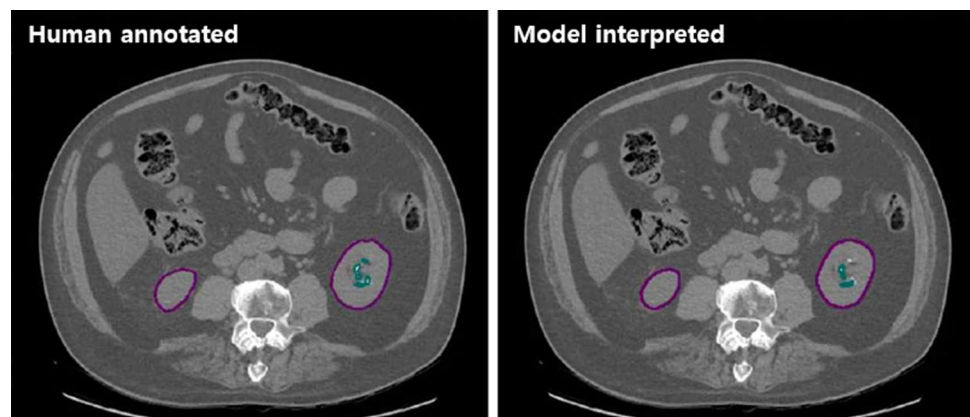


Table 2 Predictive accuracies of the development and external validation cohorts

Classification	Cohort	Stone type	Accuracy (%)	Sensitivity (%)	Specificity (%)
LR					
Binary	Development	UA vs. non-UA	82.4	72.4	69.6
	External validation	UA vs. non-UA	86.9	82.6	73.3
ML					
Binary	Development	UA vs. non-UA	99.9	100.0	99.9
	External validation	UA vs. non-UA	97.1	89.4	98.6
Multiclass	Development	UA	98.2	95.7	89.6
		Calcium oxalate	88.4	70.7	98.4
		Struvite	98.7	97.7	99.2
		Cystine	95.5	77.1	96.0
	External validation	UA	91.3	77.1	89.6
		Calcium oxalate	74.3	95.8	55.8
		Struvite	89.4	72.5	97.2
		Cystine	93.5	0.0	93.5

UA uric acid

Binary classification

The development model discriminated UA and non-UA stones with an accuracy of 99.9%, with 100.0% sensitivity and 99.9% specificity (Table 2). We identified features most predictive for binary stone classification by quantifying the predictor importance of each variable. Stone density, diabetes mellitus, and urinary pH showed to be the top three contributing features for classifying UA stones. On external validation, the model performed with a predictive accuracy of 97.1%, with 89.4% sensitivity and 98.6% specificity. The ML-based model exhibited higher accuracy compared to the multivariate logistic regression model in both development (99.9% vs. 82.4%) and external validation (97.1% vs. 86.9%) cohorts.

Multiclass classification

The development model discriminated UA, calcium oxalate, struvite, and cystine stones with predictive performances of 98.2%, 88.4%, 98.7% and 95.5%, respectively (Table 2). The features most predictive for multiclass stone classification were stone density, diabetes mellitus, and urinary pH. On external validation, the model's prediction accuracies for UA, calcium oxalate, struvite, and cystine stones were 91.3%, 74.3%, 89.4% and 93.5%, respectively. These ML-based model's accuracies in the multiclass classification were relatively lower than 99.9% and 97.1% for the binary classification in the development and external validation cohorts, respectively; however, they remained within clinically acceptable ranges.

The features of 21 (10.4%) patients that were misclassified were compared to 181 (89.6%) patients who were correctly classified in the total cohort. The level of serum potassium and urea were significantly higher in patients with misclassified stones.

Discussion

The inspiration for this study arises from the unmet clinical need to accurately predict UA stones prior to selecting the optimal treatment modality. Over the last decade, technological advancements in ureteroscopes and laser lithotriptors have paved the path to an upsurge in surgical intervention regardless of stone composition [15, 16]. Although alkalinization therapy for UA stones is ideal for patients with high morbidity or recurrent UA stone formers, it is commonly underused due to the lack of reliable factors predicting its outcome, concerns about the existence of heterogeneous stone composition, and patient intolerance [7–15]. Most of all, the lack of standardized protocols for predicting UA components adds complexity to making treatment decisions

in real-life situations [15]. The present study is the first to develop and validate an effective predictive model incorporating NCCT images into traditional demographic and clinical data to classify UA from non-UA stones in patients with stones in the 'grey zone' HUs. External validation showed that our objective, expeditious, and non-invasive model could identify UA stones with an accuracy of 97.1%, the highest predictive performance reported to date.

Our model has several implications for improving the current standard of care through its implementation in clinical practice. First, the input variables, including demographic, clinical, and NCCT data, are those readily available in real-world practice, which supports the general applicability of our model. Previously reported stone component classification models utilizing imaging data generally require time-consuming manual analysis of HU parameters or additional examination using specific CT scanner types, such as DECT, which may not be available across all practice settings [8, 17–19]. In contrast, our automated model utilizes conventional CT imaging, which may reduce the time required for classification and avoid radiation exposure and costs associated with additional CT imaging.

Second, we selected patients with stones of relatively low HUs for the model development since these stones pose a diagnostic dilemma in clinical decision-making for alkalinization therapy [5]. We selected stones with HUs < 800 in order to include struvite and cystine stones, in addition to UA stones, that are characterized as having a completely distinct management approach. The multiclass classification model provided a relatively lower performance than the binary classification. However, the overall performance was excellent and surpassed that of the conventional multivariate logistic regression model, providing a reliable diagnostic standard for treatment decision-making. Finally, the architecture of our model and its working principle allow future refinements. To date, there is no definite consensus regarding the optimal laser settings according to distinct stone features. In future studies, preoperative data obtained from our model can be integrated with intraoperative laser lithotripsy data. Such investigation may potentially provide stone-specific optimal laser settings for maximal fragmentation efficiency according to each stone feature.

Several strengths of our study are worth mentioning. First, external validation of prediction models is essential before their use in clinical practice. Since validation samples should be obtained from different but plausibly relevant cohorts, the performance of our model was validated with an external cohort comprised of patients from an international institution with distinct ethnic backgrounds. Discrimination performance is usually observed to be inferior in the external validation cohort compared to the development cohort [20]. Nevertheless, the performance of our external validation cohort was non-inferior compared to that of the

development cohort, indicating the validity and feasibility of our model. Second, we incorporated a comprehensive set of demographical, clinical, and NCCT imaging data that are potentially associated with stone components for the model development. Moreover, the dataset was considered of high quality, with all input variables of the development and external validation cohorts being manually reviewed and incorporated without any missing data, which may have contributed to its high predictive performance. Third, our model is based on a fully automated algorithm which has the capability of stone recognition and component discrimination that may minimize manual handling processes. Our model can be integrated into any electronic medical record systems that use coding algorithms to be utilized as a decision support system [21].

This study is not without limitations. First, mixed component stones were excluded from the development and external validation cohorts. Although the distinction between pure UA stones and mixed component stones is crucial in the decision-making of alkalization therapy, only stones with pure components were included. Since the extent of the UA component beyond which the stone has to be defined as mixed is unclear, subsequent studies incorporating quantitative analysis of mixed stones, will need to be performed to screen optimal patients who would be amenable to medical therapy. Second, a population-based database with a larger number of subjects may provide better generalizability. Albeit, we utilized institutional data, which provided a comprehensive and high-quality dataset, to maximize predictive performance. Third, NCCT scans were performed at various referral centers; and therefore, there was no standardization of scan setting. Finally, performances declined for the multiclass classification, indicating uncertainty of clinical usefulness, especially in classifying cystine stones. The likely explanation is the limited number of cystine stones in both the development and external validation cohorts. Subsequent studies will be needed to address other reasons underlying the misclassification and to refine the accuracy of our model.

Notwithstanding these limitations, the advantages of our model over previously reported tools classifying predicting stone components indicate its feasibility and general applicability to be implemented into real-world clinical practice.

Conclusions

We developed and externally validated an ML-based model to identify and delineate kidney stones and classify UA stones from other stone components. With the highest predictive performance reported to date, our model can be reliably used to select candidates for an earlier-directed alkalization therapy in patients with kidney stones within the 'grey zone' HUs.

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Author contributions Conceptualization: KCK and BHC; methodology: VKFW and AH; formal analysis and investigation: SL, SB, and HK; writing—original draft preparation: KCK; writing—review and editing: VKFW, AH, SL, SB, and HK; funding acquisition: KCK; resources: BHC; supervision: KCK and BHC.

Data availability The data sets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest No competing financial interests exist for all authors, including Ben H. Chew, Victor KF. Wong, Abdulghafour Hala-wani, Sujin Lee, Sangyeop Baek, Hoyong Kang, and Kyo Chul Koo.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

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